

Comprehending and modifying ER diagrams and UML class diagrams

TASK: T1

TREATEMENT: CD

First Name:.....

Last Name:.....

ID:.....

Start time:

Figure 1 shows the UML class diagram of a system managing the hospital activities. In particular, the diagram regards the patients, doctors, examinations, and laboratory information.

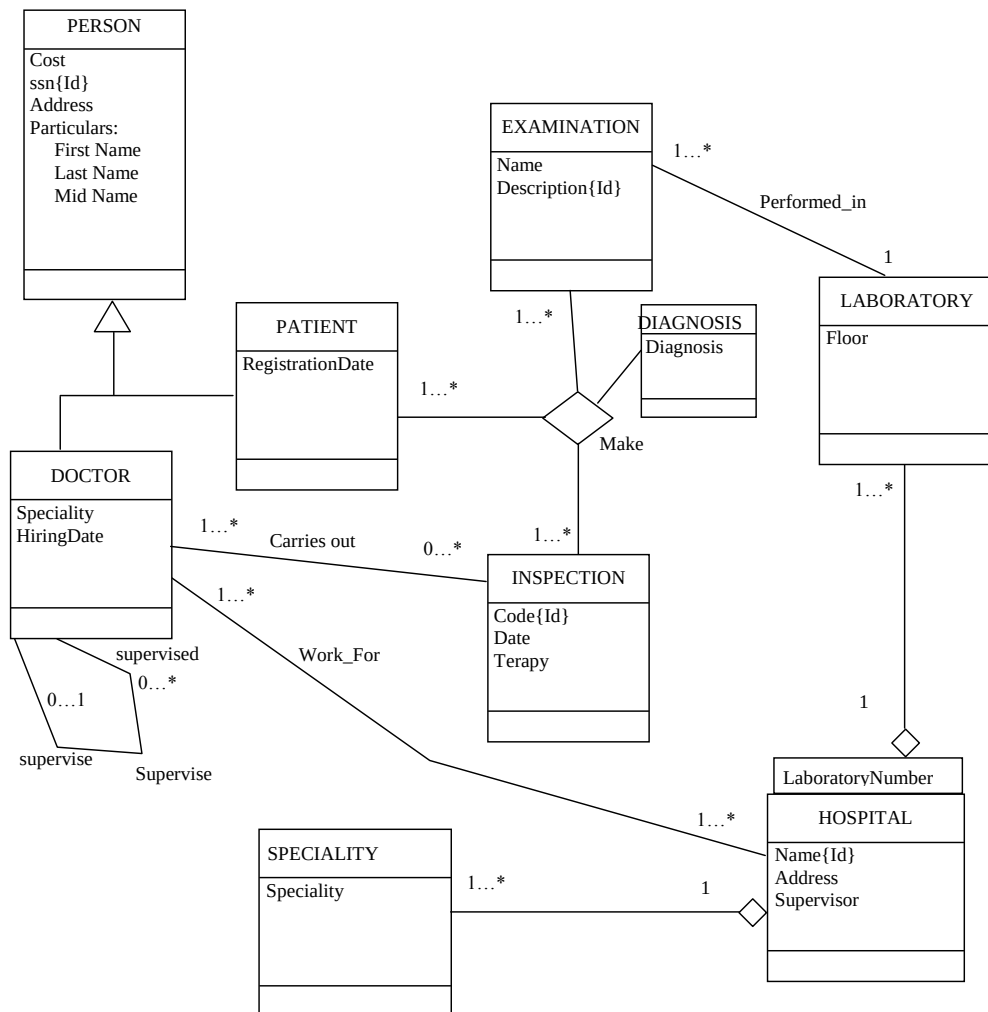
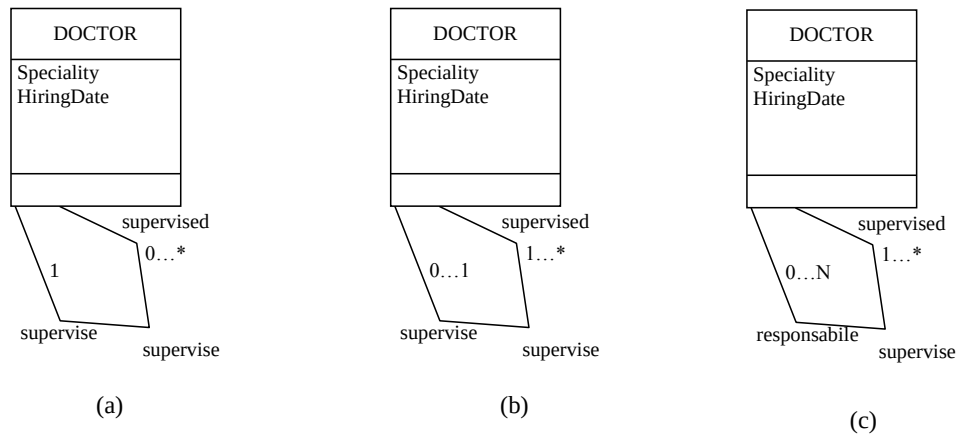


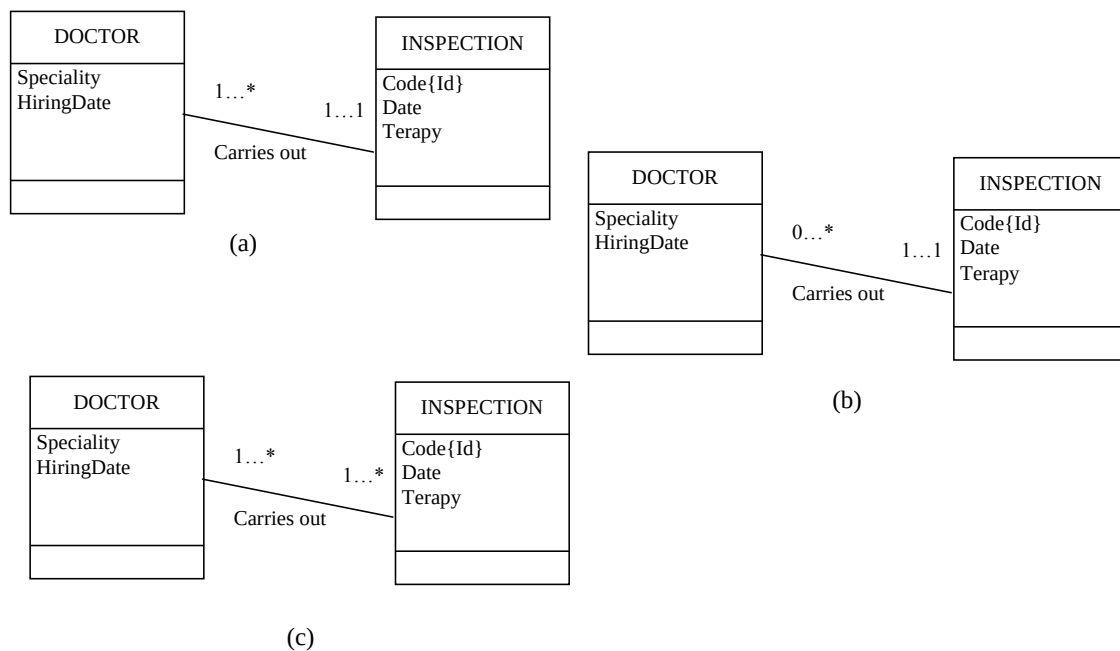
Figura 1: UML class diagram of a system managing hospital information

Analyze the diagram in Figure 1 and answer the following questions, about comprehension and modification activities.

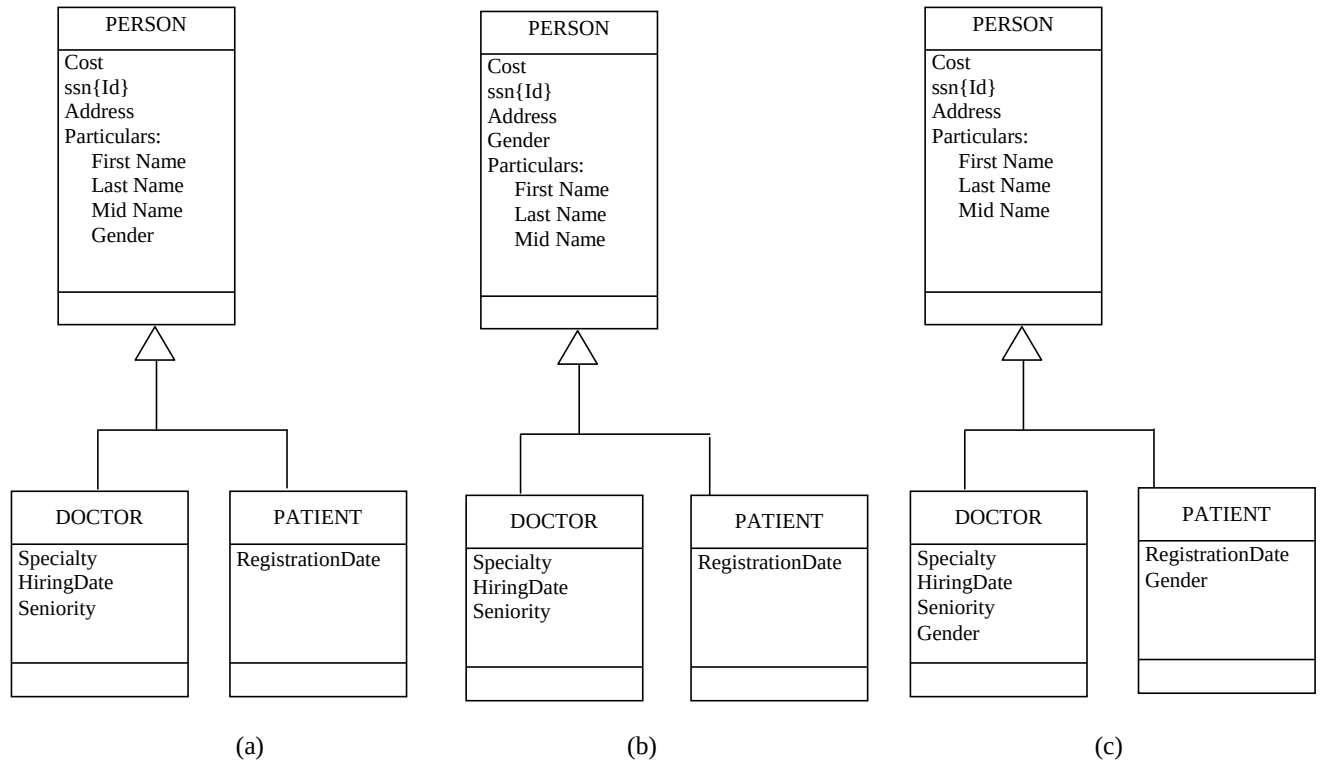
Q1 We want specify a supervisor for each doctor. A doctor can be supervisor of several doctors. Indicate the part of the class diagram interested by this modification activity.



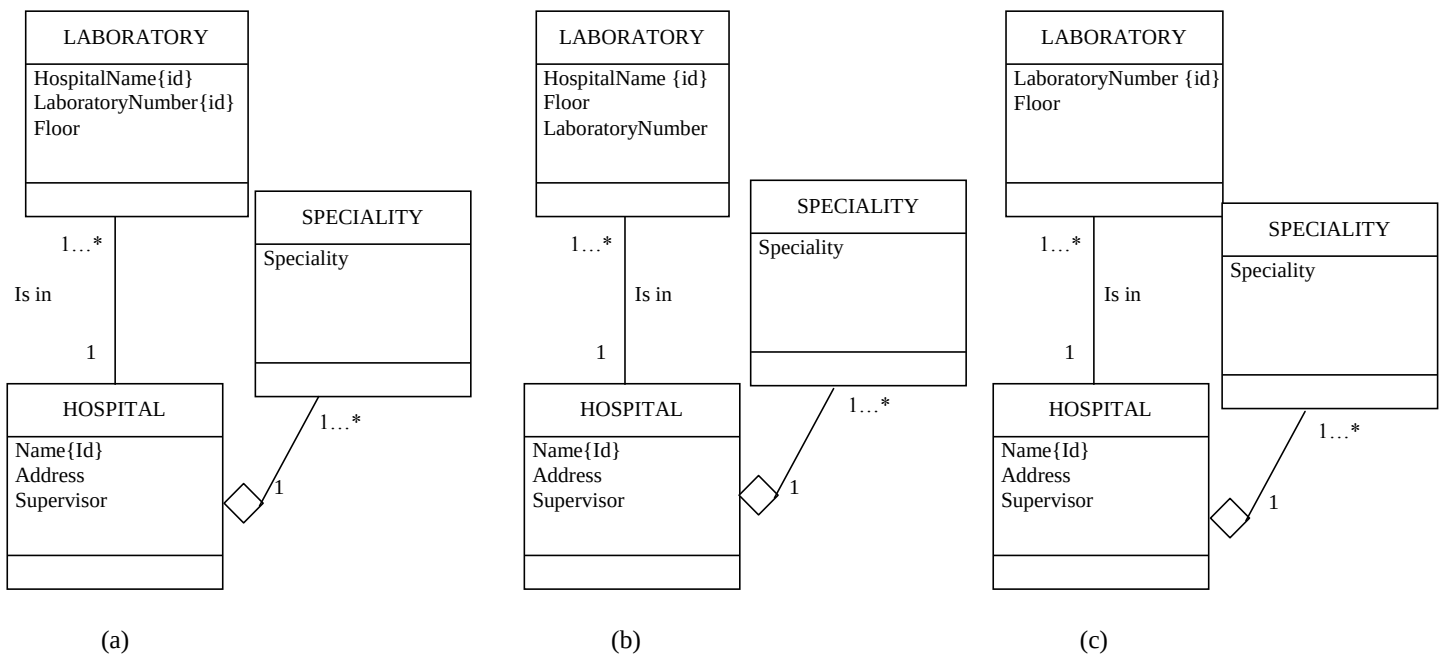
Q2 We want specify that a doctor carries out at least an inspection. An inspection can be carried out by several doctors. Indicate the part of the class diagram interested by this modification activity.



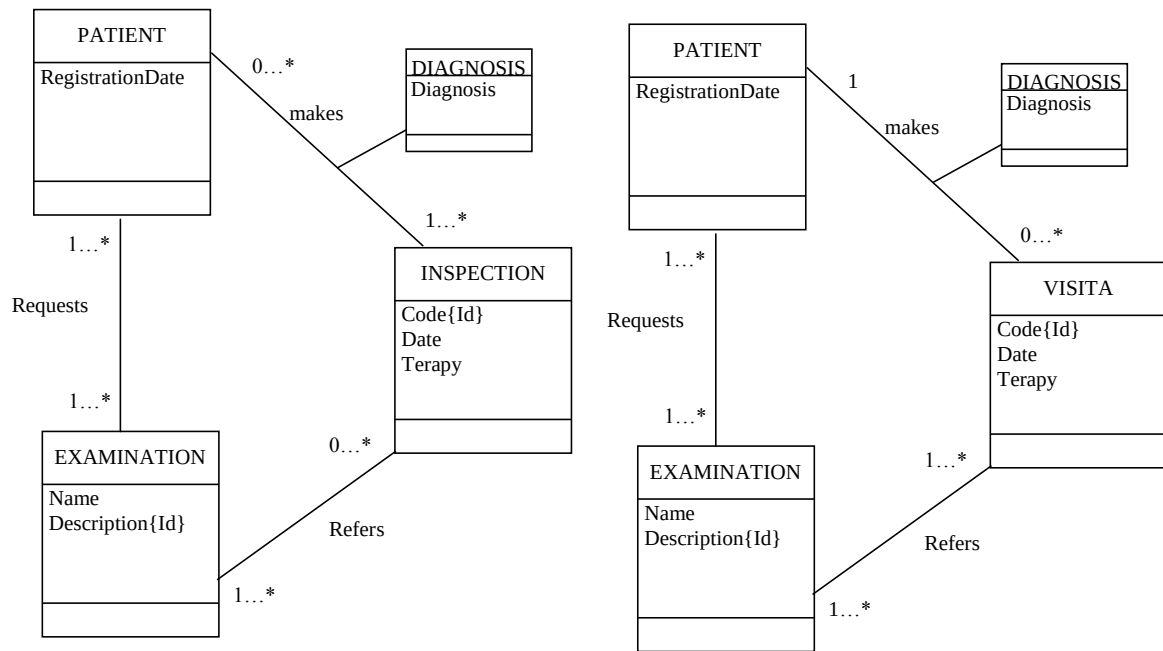
Q3 We want to add to the particulars of persons the information on gender (female/male). Indicate the part of the class diagram interested by this modification activity:



Q4 We want specify that Laboratory is not a weak entity. Indicate the part of the class diagram interested by this modification activity:

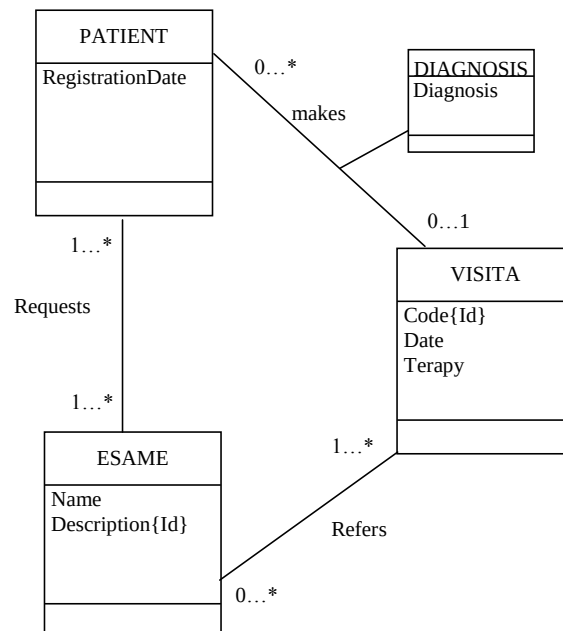


Q5 We want specify that a patient can make at least an examination without an inspection. A patient can make several examination and can do several inspections. Furthermore, an inspection is made by only one patient, but it can refer to several examinations (It regards at least an examination). An examination can be made by several patients and can refer to several inspections Indicate the part of the class diagram interested by this modification activity:



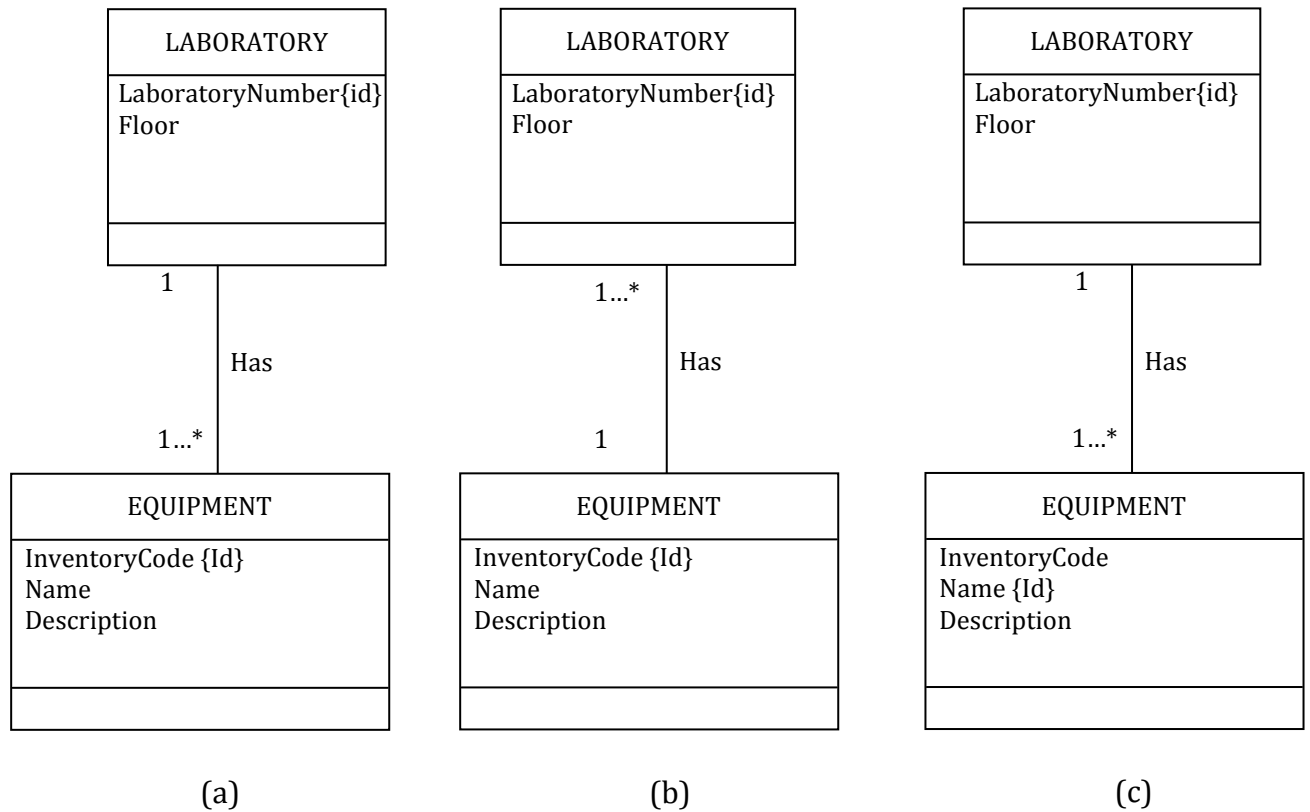
(a)

(b)

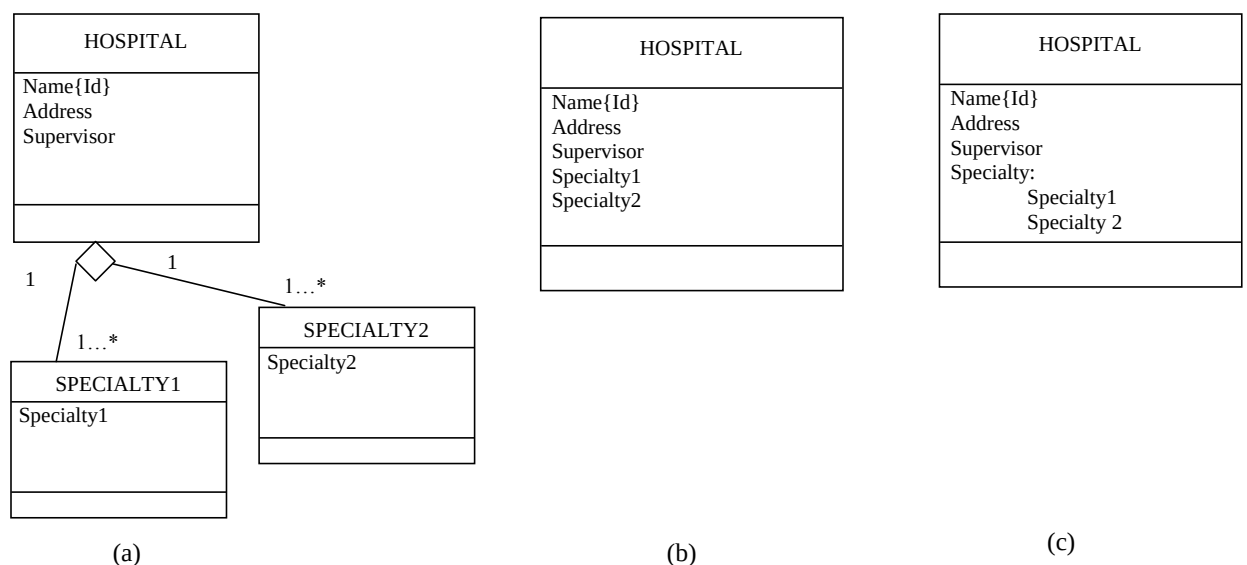


(c)

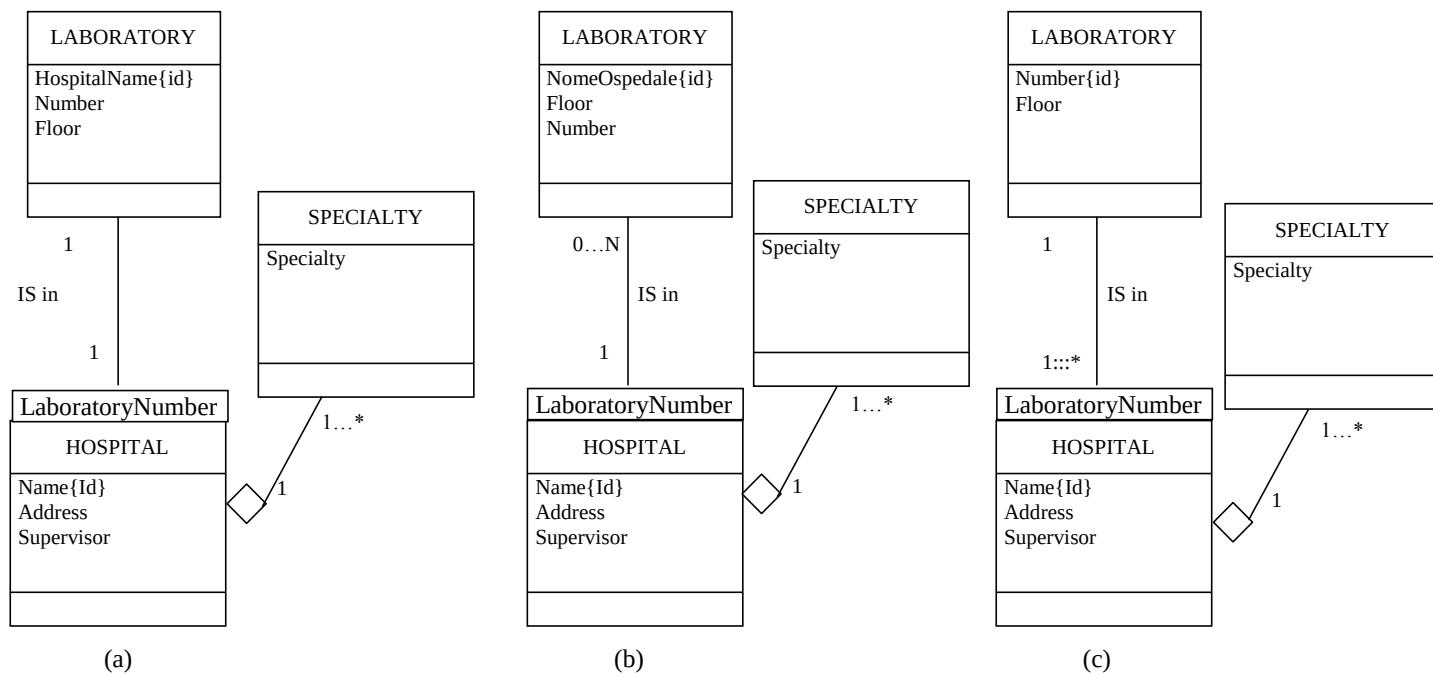
Q6 We want to specify the information about the equipments present in each laboratory. A laboratory can contain several equipments and each equipment is identified by inventory code. A description and a name have also to be specified for an equipment. Indicate the part of the class diagram interested by this modification activity:



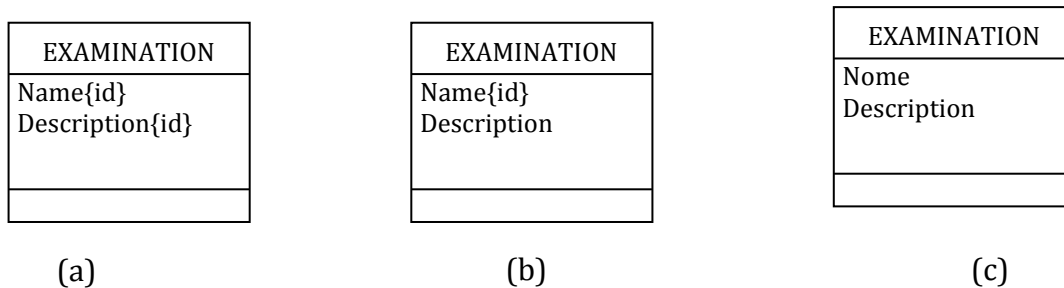
Q7 We want specify that each hospital has at most two specialties: **Specialty1** and **Specialty2**. Indicate the part of the class diagram interested by this modification activity:



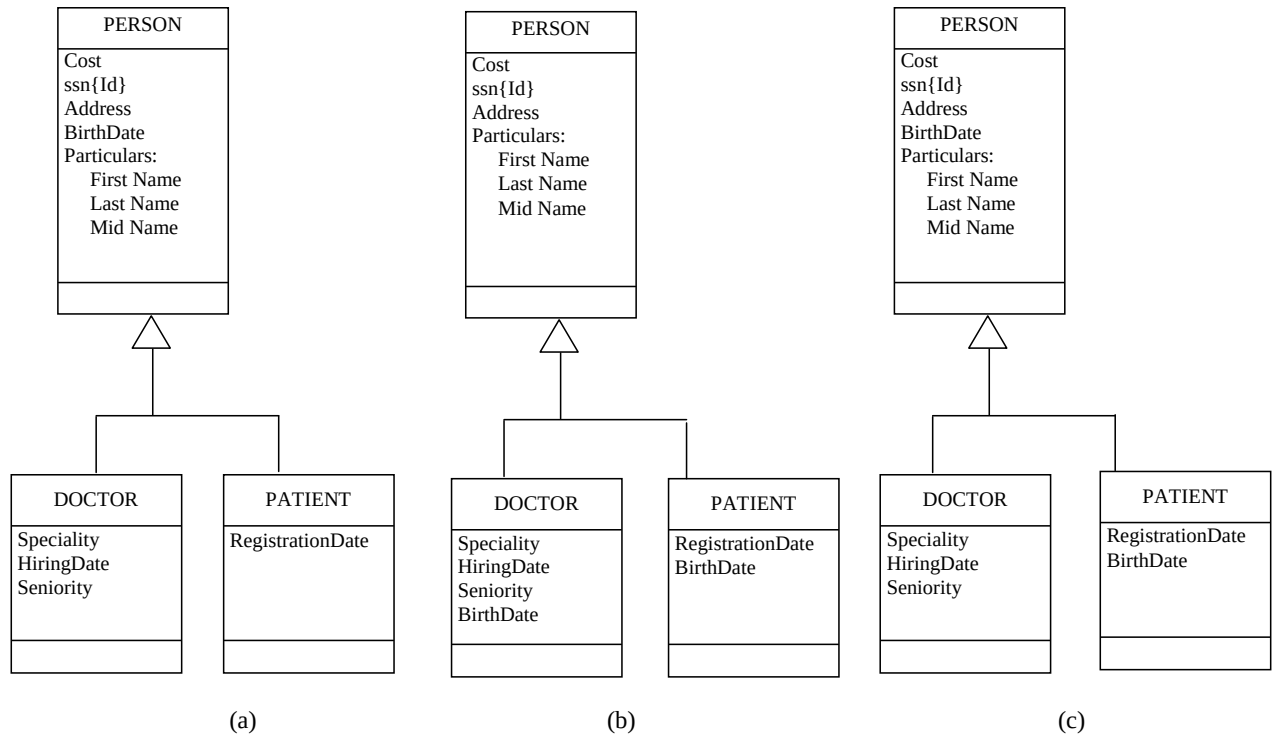
Q8 We want to specify that each hospital has one laboratory in. Indicate the part of the class diagram interested by this modification activity:



Q9 We want to identify each examination by its name. Indicate the part of the class diagram interested by this modification activity:



Q10 We want to add the birth date of each patient. Furthermore, for each doctor we want to specify the birth date and the seniority. Indicate the part of the class diagram interested by this modification activity:



End time:

Final questionnaire

The following questions have to answered after the execution of the comprehension and modification task

A1 I had enough time to perform the tasks

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A2. The task objectives were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A3. The tasks I performed were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A4. Judge the difficulty of the comprehension and modification task

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree